

RAG Test Knowledge Document

This document is intentionally structured with clear headings, definitions, facts, and explanations so it can be indexed and queried by Retrieval-Augmented Generation (RAG) systems. It contains multiple technical topics including Python, Web Development, Databases, Operating Systems, Networking, and Artificial Intelligence.

Python Programming

Python is a high-level, interpreted programming language created by Guido van Rossum and first released in 1991.

Python emphasizes readability and uses indentation instead of braces to define code blocks.

Common data types in Python include integers, floats, strings, lists, tuples, sets, and dictionaries.

A list in Python is ordered and mutable, while a tuple is ordered but immutable.

A dictionary stores key-value pairs and allows fast lookups using hashing.

Python supports multiple programming paradigms including procedural, object-oriented, and functional programming.

The pip tool is used to install external Python packages from the Python Package Index (PyPI).

Virtual environments isolate project dependencies and are created using `'python -m venv env'`.

Flask and Django are two popular Python web frameworks.

Flask Web Framework

Flask is a lightweight web framework written in Python.

It is based on the WSGI toolkit Werkzeug and the template engine Jinja2.

Flask is considered a micro-framework because it does not require particular tools or libraries.

A route in Flask is created using the `@app.route` decorator.

The request object represents incoming client data such as form input or query parameters.

The `render_template` function is used to render HTML templates stored in the templates folder.

Flask is commonly used for APIs, small web apps, and prototypes.

Flask applications typically run on localhost port 5000 in development mode.

Database Fundamentals

A database is an organized collection of structured information or data.

Relational databases store data in tables consisting of rows and columns.

Primary keys uniquely identify each record in a table.

Foreign keys create relationships between tables.

SQL stands for Structured Query Language and is used to interact with relational databases.

Common SQL commands include SELECT, INSERT, UPDATE, and DELETE.

Normalization is the process of organizing a database to reduce redundancy.

Examples of relational databases include MySQL, PostgreSQL, and SQLite.

Operating Systems

An operating system manages computer hardware and software resources.

It acts as an intermediary between users and hardware.

Process management handles creation, scheduling, and termination of processes.

Memory management allocates RAM to running programs.

A file system organizes files on storage devices.

Linux is an open-source operating system based on the Unix architecture.

Popular Linux distributions include Ubuntu, Arch Linux, and Fedora.

A kernel is the core component of the operating system responsible for hardware interaction.

Computer Networking

A computer network is a group of interconnected devices that communicate with each other.

The Internet is the largest computer network in the world.

IP stands for Internet Protocol and identifies devices using IP addresses.

IPv4 addresses are 32-bit numbers while IPv6 addresses are 128-bit.

HTTP stands for HyperText Transfer Protocol and is used for web communication.

HTTPS is a secure version of HTTP using encryption via TLS/SSL.

DNS translates domain names into IP addresses.

Routers direct packets between networks while switches connect devices within a local network.

Artificial Intelligence Basics

Artificial Intelligence (AI) refers to machines simulating human intelligence.

Machine Learning is a subset of AI that learns patterns from data.

Supervised learning uses labeled datasets.

Unsupervised learning finds patterns in unlabeled data.

Neural networks are inspired by the human brain and consist of layers of interconnected nodes.

Natural Language Processing (NLP) allows computers to understand human language.

Examples of AI applications include recommendation systems, chatbots, and image recognition.

Large Language Models (LLMs) are trained on massive text datasets to generate human-like responses.

Glossary

Term	Definition
Kernel	Core part of an operating system managing hardware
SQL	Language used to query relational databases
HTTP	Protocol used for transferring web pages
API	Interface allowing software applications to communicate
DNS	System translating domain names to IP addresses
Virtual Environment	Isolated Python dependency environment

Frequently Asked Questions

Q: What is RAG?

A: Retrieval-Augmented Generation is a technique where a language model retrieves external documents and uses them to answer questions.

Q: What is a primary key?

A: A primary key uniquely identifies each record in a database table.

Q: What is Flask mainly used for?

A: Flask is commonly used to build APIs and lightweight web applications.

Q: What does DNS do?

A: DNS converts human-readable domain names into IP addresses.

Q: What is a neural network?

A: A neural network is a computational model composed of layers of interconnected nodes that learn patterns from data.